SOC Advanced Operations Report

Topics: Advanced Log Analysis, Threat Intelligence Integration, Incident Escalation Workflows

1. Advanced Log Analysis

Advanced log analysis enables SOC analysts to detect complex cyber threats by correlating logs from multiple systems including firewalls, operating systems, applications, and endpoint security tools. By analyzing logs together, analysts can identify patterns that indicate malicious activity.

Log Source	Event	Observation
Windows Event Log	Event ID 4625	Multiple failed login attempts
Firewall Logs	Outbound connection	Suspicious external IP communication
Endpoint Logs	Process execution	Unknown process started

Correlation of these logs may indicate brute■force login attempts followed by unauthorized access. Anomaly detection techniques also help identify unusual behaviors such as abnormal login times, large data transfers, or sudden spikes in network activity.

2. Threat Intelligence Integration

Threat intelligence provides contextual information about known cyber threats. It includes indicators of compromise (IOCs) such as malicious IP addresses, domain names, file hashes, and attacker techniques known as tactics, techniques, and procedures (TTPs). Integrating threat intelligence feeds into SIEM platforms helps SOC teams detect threats faster.

Indicator Type	Example	Purpose
Malicious IP	192.168.1.200	Detect command and control servers
File Hash	SHA256 hash	Identify malware samples
Domain	malicious-domain.com	Detect phishing infrastructure

SOC analysts can also perform threat hunting using intelligence frameworks such as MITRE ATT&CK; to proactively search for attacker techniques like misuse of valid accounts or lateral movement.

3. Incident Escalation Workflows

Incident escalation workflows ensure that security incidents are handled by the appropriate level of expertise within a SOC team. Most SOC environments operate with a tiered structure where each level handles increasing complexity.

SOC Tier	Role	Responsibilities
Tier 1	Alert Triage	Monitor alerts and perform initial investigation
Tier 2	Incident Investigation	Analyze threats and perform containment
Tier 3	Advanced Analysis	Threat hunting and malware analysis

During escalation, analysts communicate using structured reports such as Situation Reports (SITREP). Automation platforms known as SOAR tools can also automate ticket assignment, alert enrichment, and escalation processes to improve SOC efficiency.

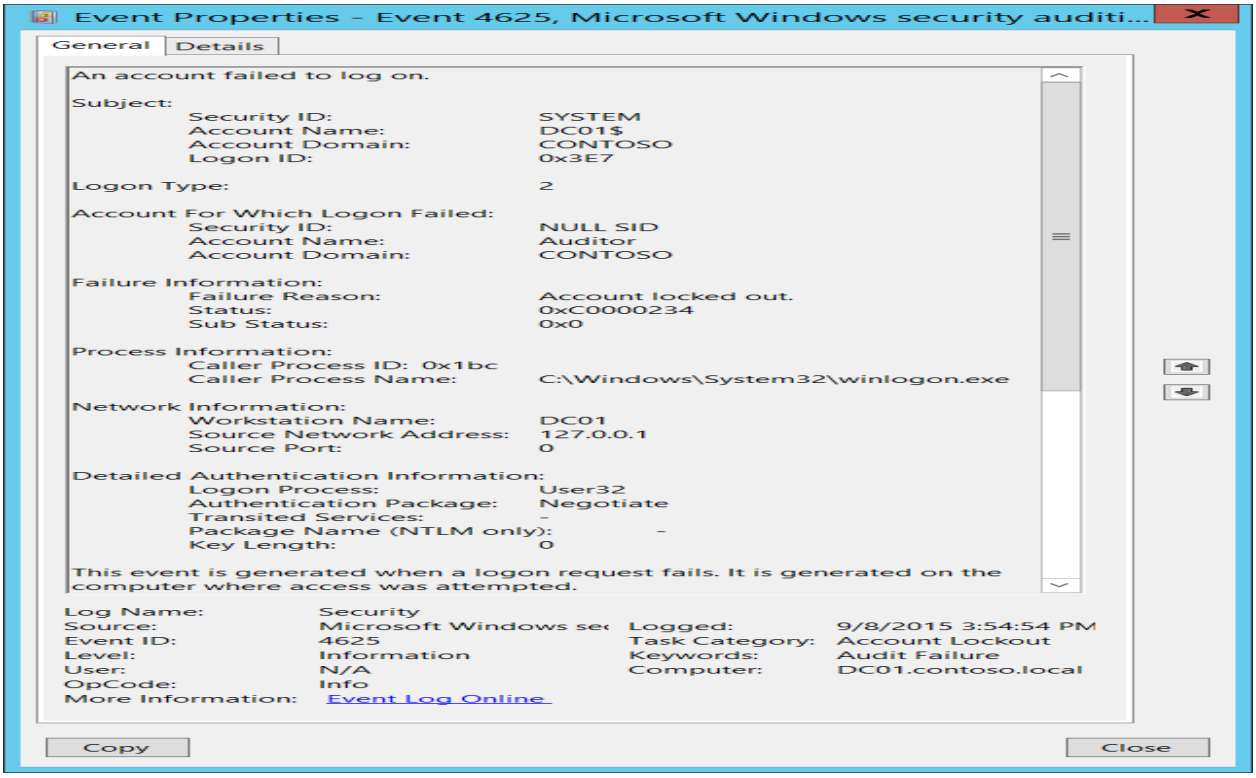
Conclusion

Advanced log analysis, threat intelligence integration, and structured escalation workflows are essential components of modern SOC operations. These capabilities enable organizations to detect complex threats, respond effectively, and continuously improve their cybersecurity posture.

SOC Practical Report: Advanced Log Analysis & Threat Intelligence

This report demonstrates practical Security Operations Center (SOC) tasks including log analysis, threat intelligence integration, anomaly detection, incident management, and automation workflows. The following screenshots and analysis show how SOC analysts investigate and respond to potential threats.

Figure 1: Security Monitoring Screenshot



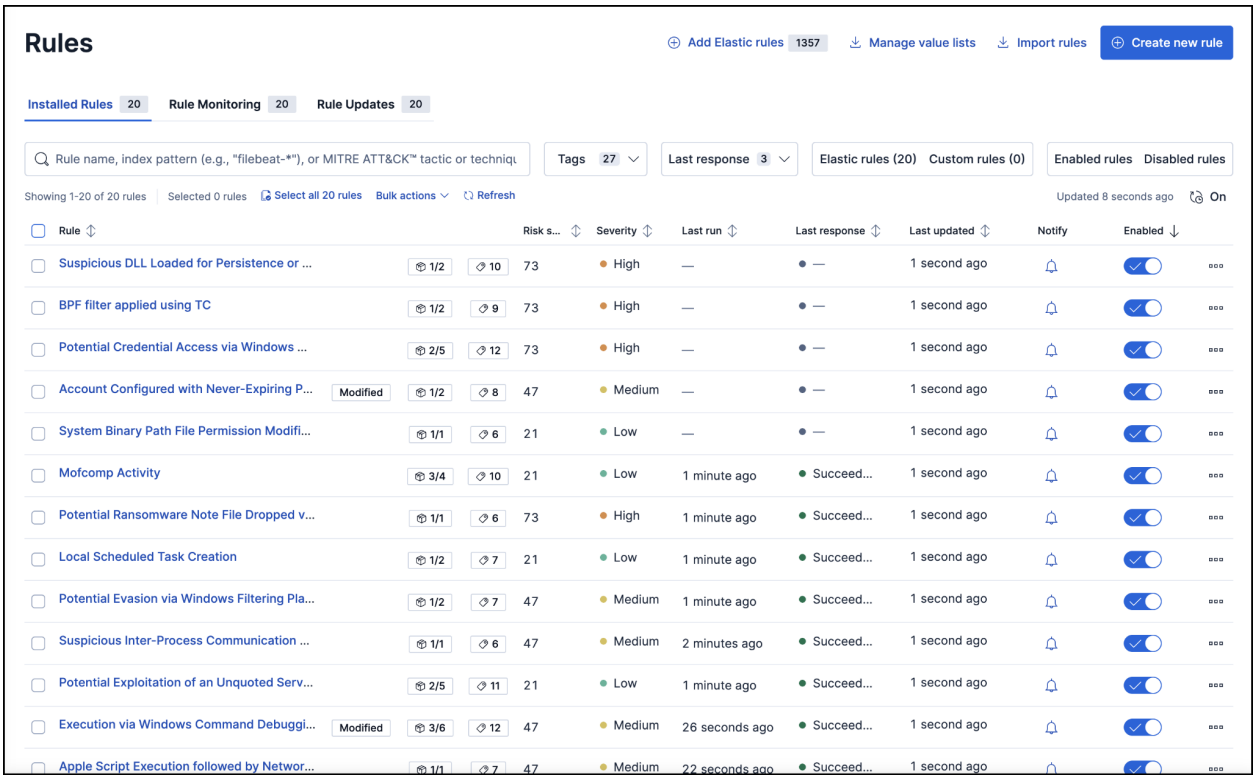
Windows Event ID 4625 shows a failed login attempt. Repeated failures may indicate brute-force attacks or incorrect credential usage.

Figure 2: Security Monitoring Screenshot

```
An account failed to log on.

Subject:
        Security ID:                    NULL SID
        Account Name:                   -
        Account Domain:                 -
        Logon ID:                       0x0

Logon Type:                             3

Account For Which Logon Failed:
        Security ID:                    NULL SID
        Account Name:                   [redacted]
        Account Domain:                 [redacted]

Failure Information:
        Failure Reason:                 Unknown user name or bad password.
        Status:                         0xC000006D
        Sub Status:                     0xC0000064

Process Information:
        Caller Process ID:  0x0
        Caller Process Name:            -

Network Information:
        Workstation Name:               [redacted]
        Source Network Address:         [redacted]
        Source Port:                    [redacted]

Detailed Authentication Information:
        Logon Process:                  NtLmSsp
        Authentication Package:         NTLM
        Transited Services:             -
        Package Name (NTLM only):       -
        Key Length:                     0
```

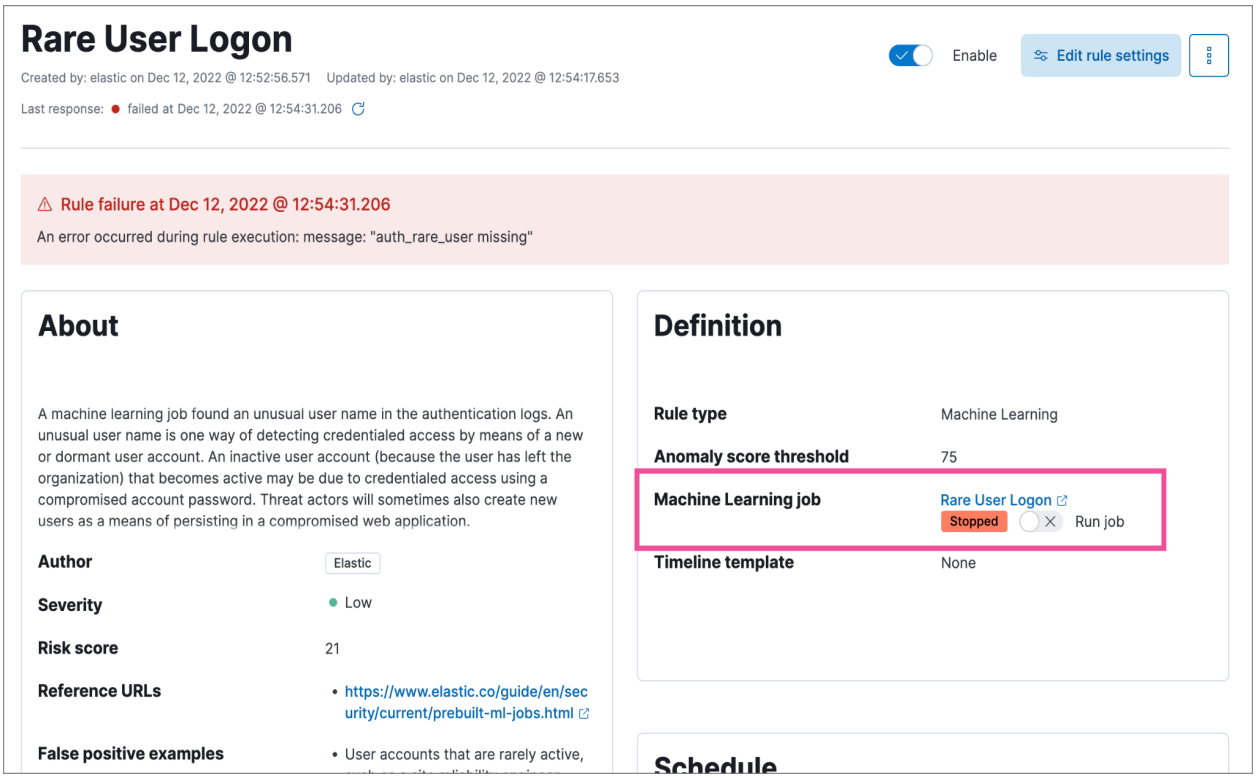
Another authentication failure event showing 'Unknown username or bad password'. Logon type 3 indicates a network login attempt.

Figure 3: Security Monitoring Screenshot



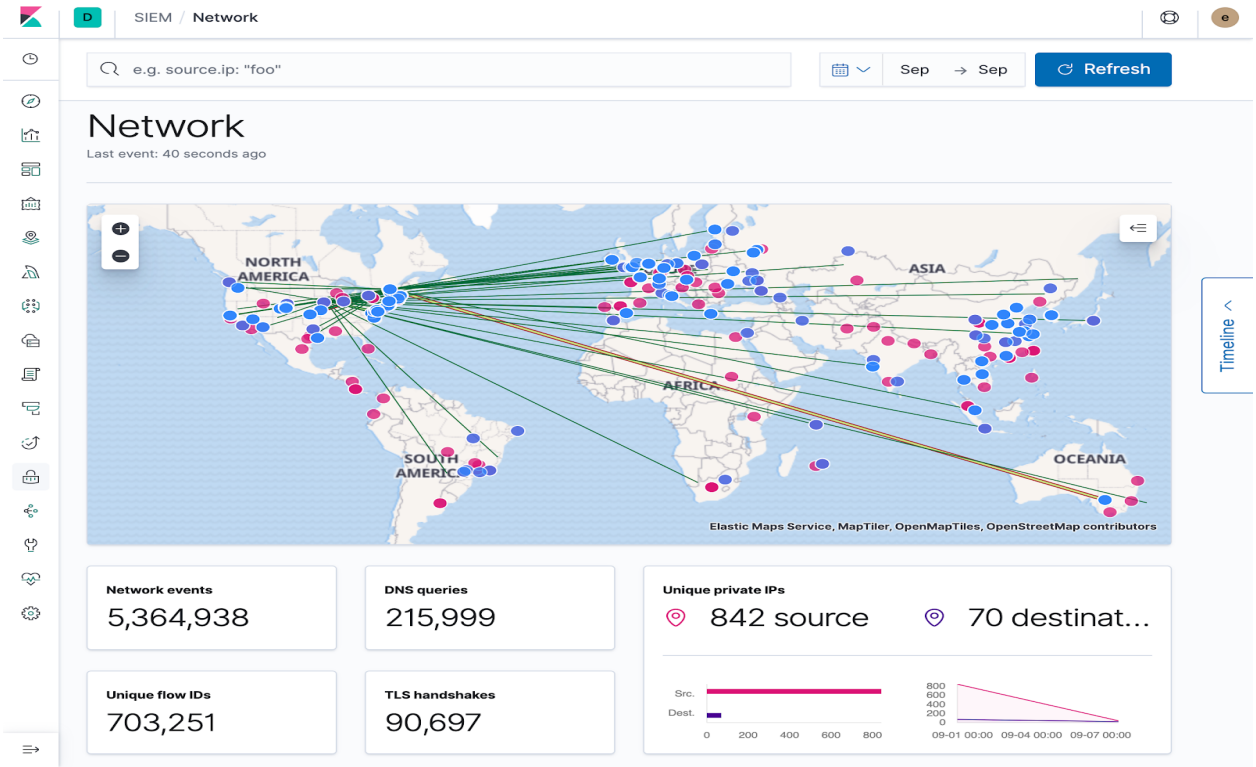
Elastic Security detection rules dashboard showing automated threat detection rules with severity levels.

Figure 4: Security Monitoring Screenshot



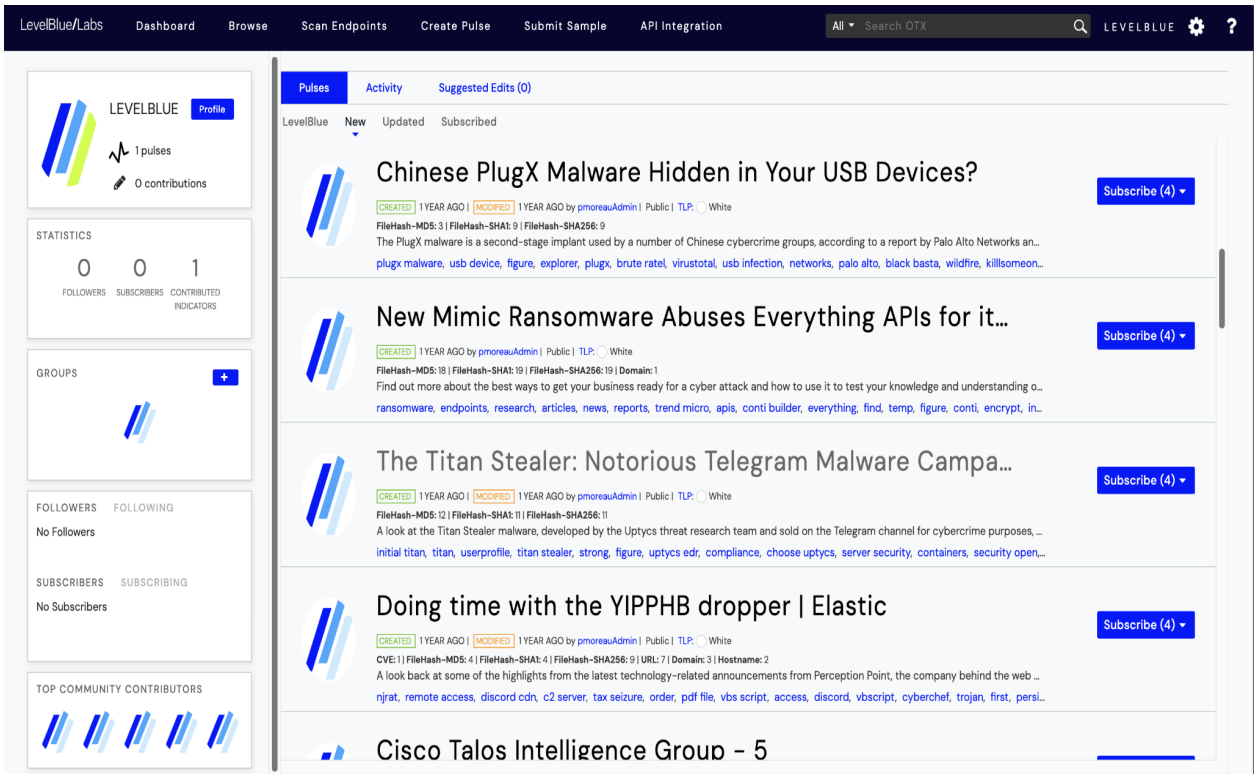
Elastic Machine Learning rule 'Rare User Logon' used to detect abnormal login behavior.

Figure 5: Security Monitoring Screenshot



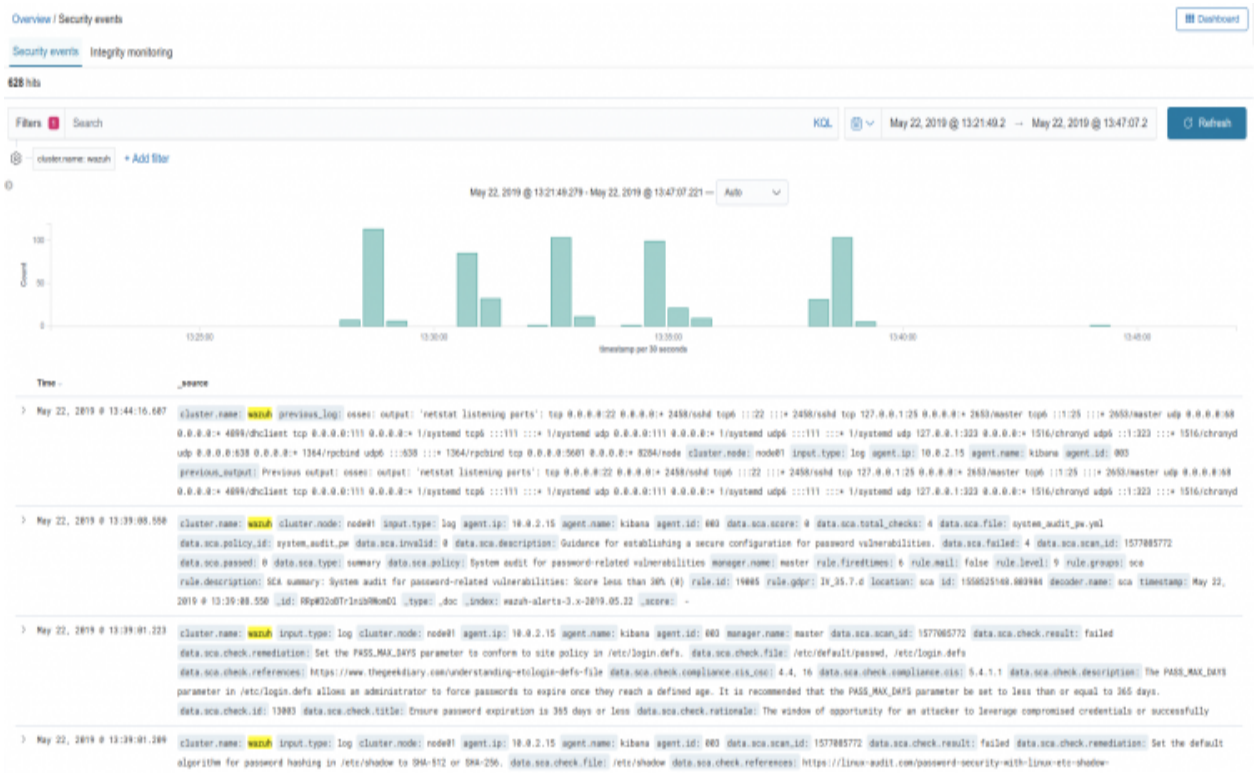
Elastic SIEM network map showing global network activity and GeoIP enrichment.

Figure 6: Security Monitoring Screenshot



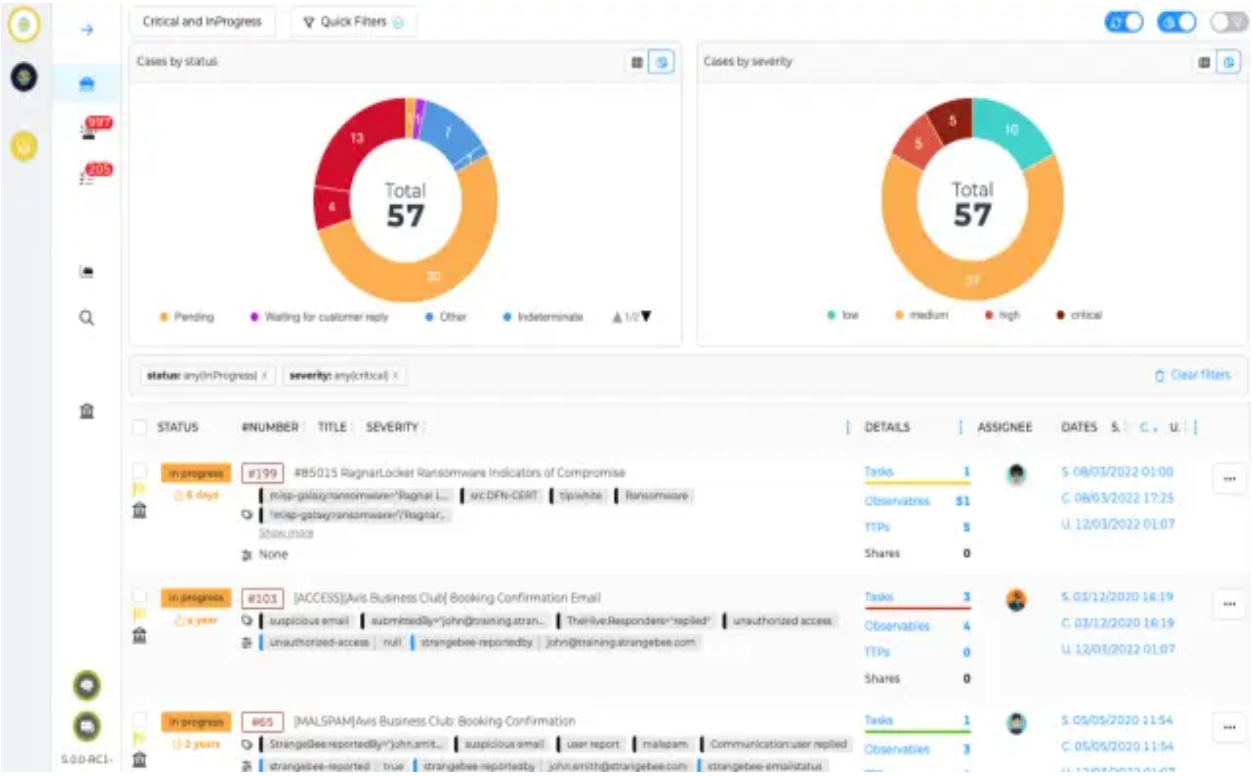
AlienVault OTX threat intelligence dashboard showing threat pulses and malware intelligence.

Figure 7: Security Monitoring Screenshot



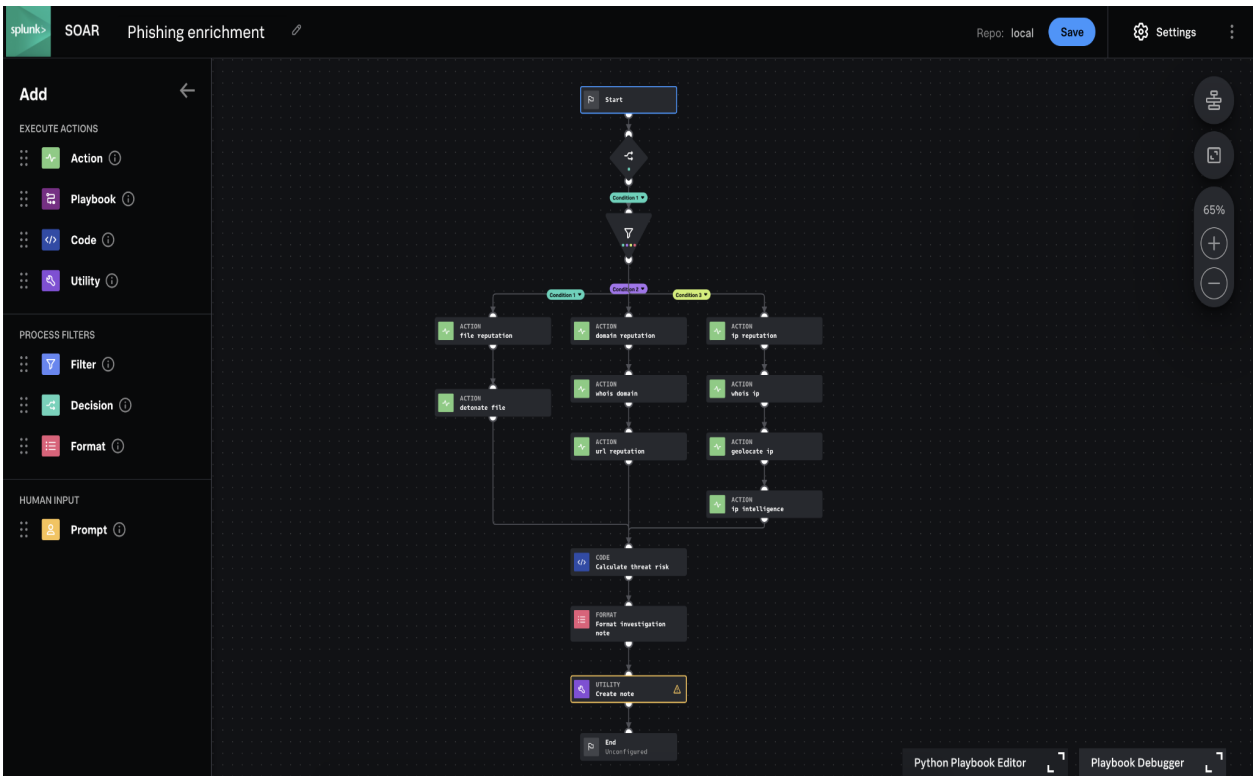
Security event monitoring logs used for correlation and investigation.

Figure 8: Security Monitoring Screenshot



TheHive incident response dashboard used to track and manage security cases.

Figure 9: Security Monitoring Screenshot



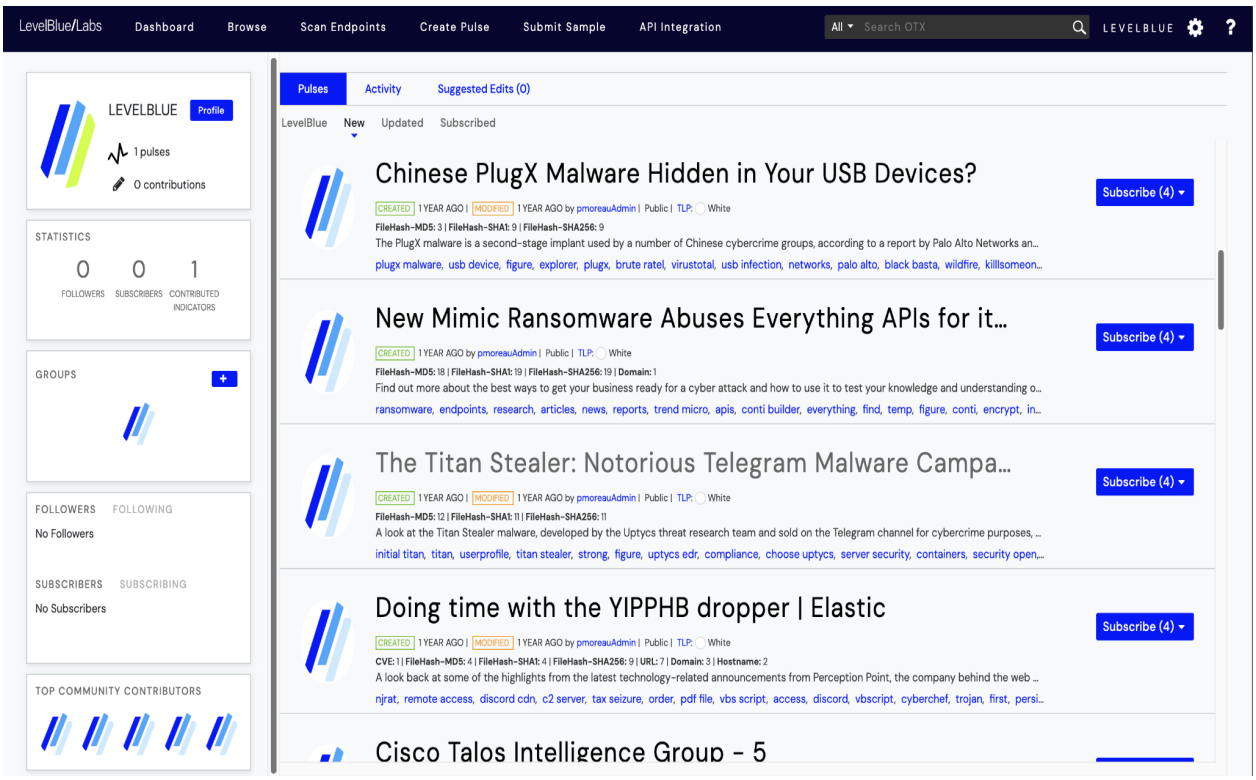
Splunk SOAR playbook showing automated threat investigation workflow.

Figure 10: Security Monitoring Screenshot

```
File  Edit  View  Search  Terminal  Help
  GNU nano 4.8                                   local_rules.xml
<!-- Local rules -->

<!-- Modify it at your will. -->
<!-- Copyright (C) 2015, Wazuh Inc. -->

<!-- Example -->
<group name="local,syslog,sshd,">

  <!--
  Dec 10 01:02:02 host sshd[1234]: Failed none for root from 1.1.1.1 port 1066 ssh2
  -->
  <rule id="100001" level="5">
    <if_sid>5716</if_sid>
    <srcip>1.1.1.1</srcip>
    <description>sshd: authentication failed from IP 1.1.1.1.</description>
    <group>authentication_failed,pci_dss_10.2.4,pci_dss_10.2.5,</group>
  </rule>

</group>

<group name="audit">
  <rule id="100002" level="3">
      <if_sid>80790</if_sid>
      <regex>downloads|tmp|temp</regex>
      <description>
          Audit: $(audit.exe) created a file with filename $(audit.file.name) in the folder $(audit.directory.name).
      </description>
      <group>audit_watch_write</group>
  </rule>
                                                  [ Read 30 lines ]
^G Get Help    ^O Write Out   ^W Where Is    ^K Cut Text    ^J Justify     ^C Cur Pos     M-U Undo       M-A Mark Text  M-] To Bracket
^X Exit        ^R Read File   ^\ Replace     ^U Paste Text  ^T To Spell    ^_ Go To Line  M-E Redo       M-6 Copy Text  ^Q Where Was
```

Wazuh custom rule configuration in local_rules.xml for detecting specific events.

Figure 11: Security Monitoring Screenshot



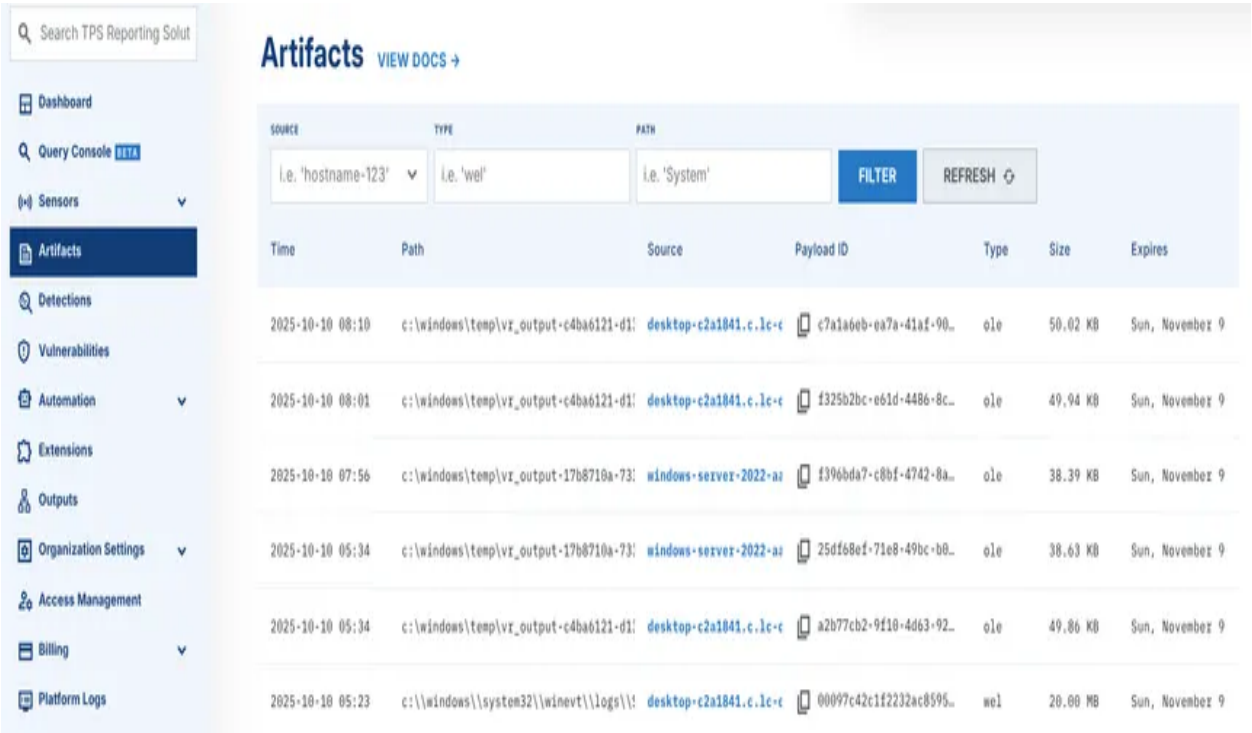
Threat intelligence and monitoring integration example used in SOC investigations.

Conclusion

The practical tasks demonstrate how SOC analysts correlate logs, detect anomalies, integrate threat intelligence, and automate response processes. Using SIEM platforms, threat intelligence feeds, and SOAR tools helps organizations detect threats faster and respond efficiently to cybersecurity incidents.

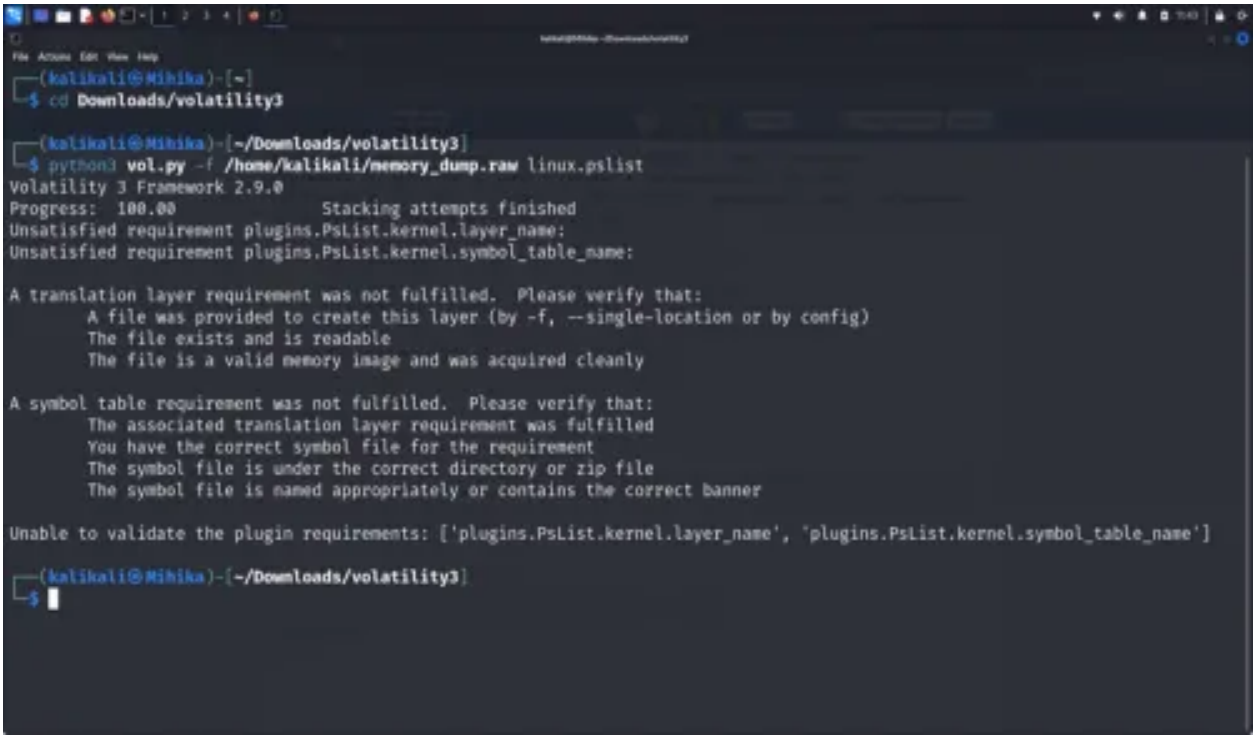
SOC Practical Report – Evidence Preservation and Security Analysis

Figure 1: Velociraptor Artifact Collection



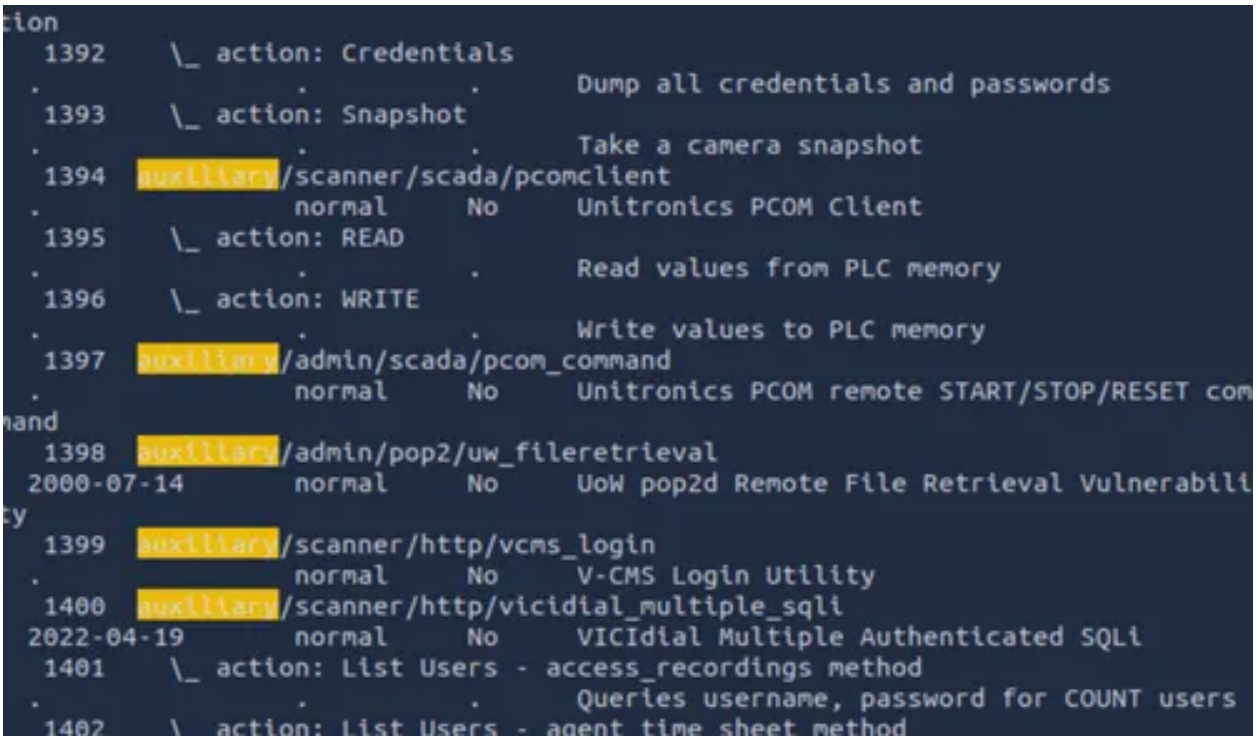
Analysis: This screenshot shows the Velociraptor artifact collection interface used in digital forensics. Artifacts such as system logs, temporary files, and endpoint data are collected from remote machines. SOC analysts use this tool to gather forensic evidence from compromised systems while maintaining evidence integrity.

Figure 2: Memory Analysis using Volatility



Analysis: This screenshot displays memory forensics performed using the Volatility framework. The command attempts to analyze a memory dump and extract running processes from system memory. Memory analysis helps identify hidden malware, injected processes, and suspicious activities that are not visible in normal logs.

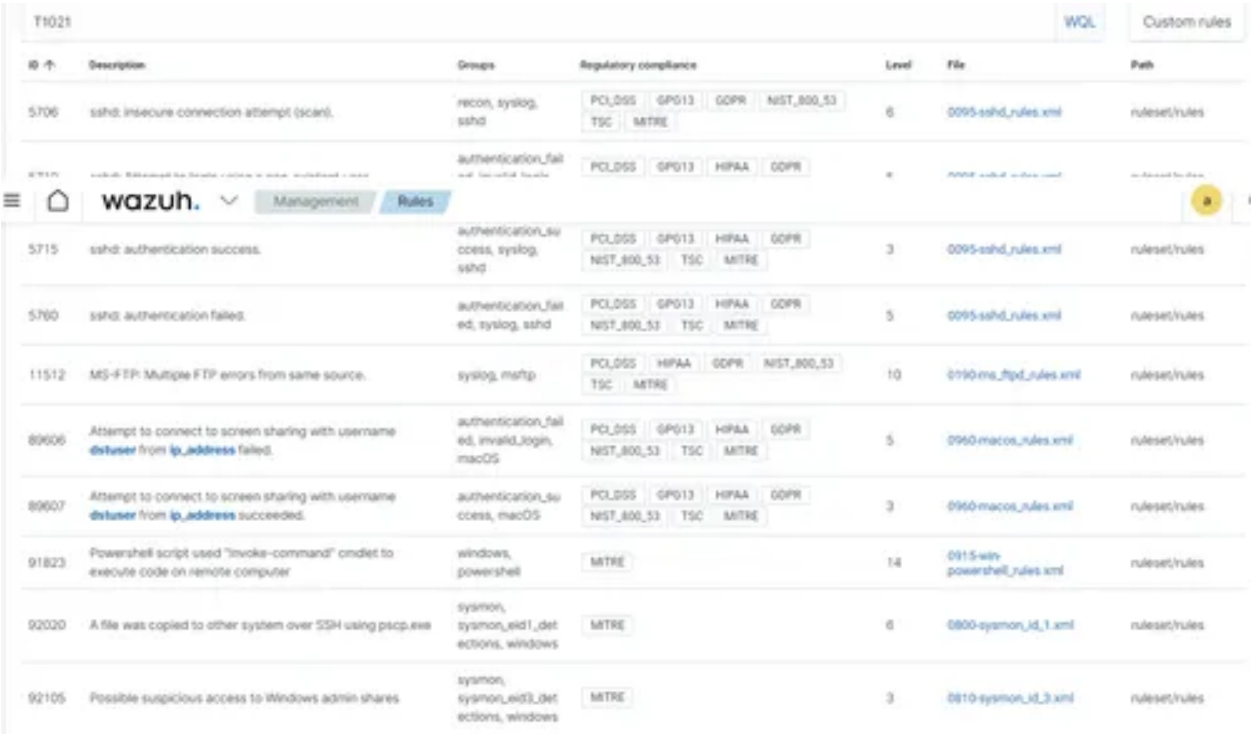
Figure 3: Vulnerability Enumeration Output



Analysis: This screenshot shows vulnerability enumeration results listing various potential vulnerabilities and services discovered during security analysis. Enumeration helps identify

weaknesses such as credential exposure, remote access vulnerabilities, and insecure services.

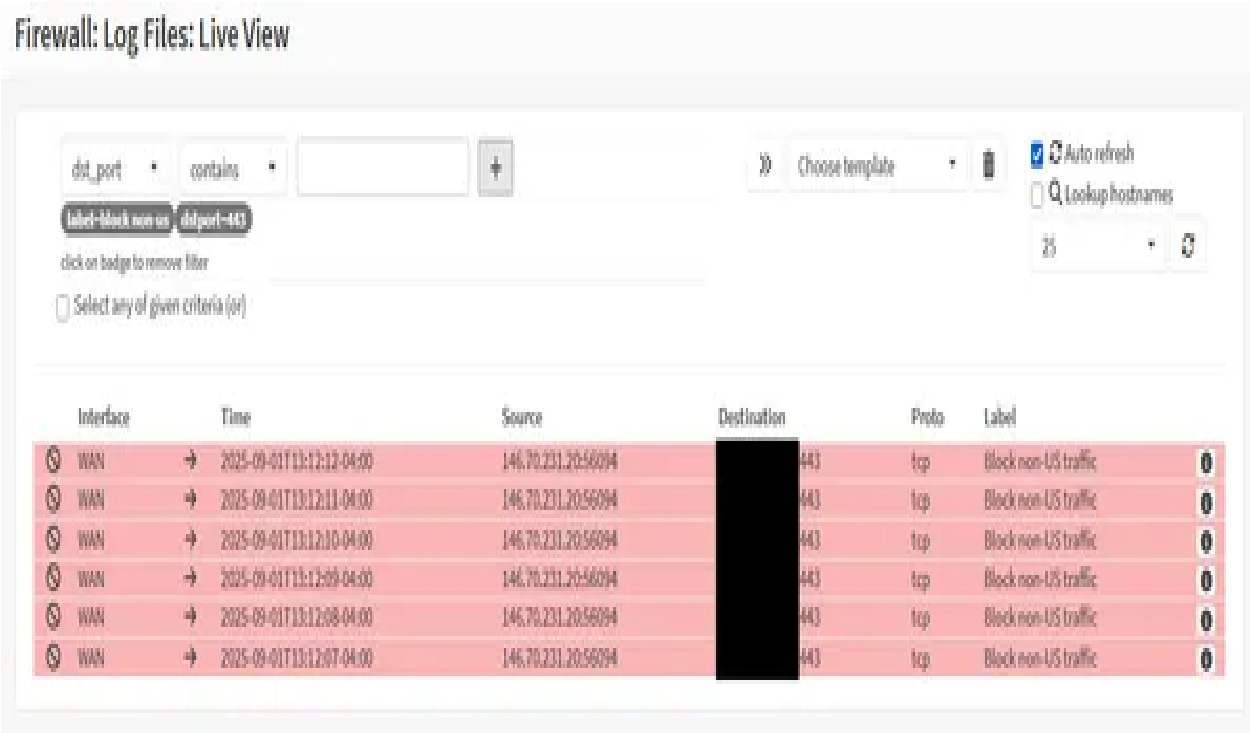
Figure 4: Wazuh Detection Rules Dashboard



ID	Description	Groups	Regulatory compliance	Level	File	Path
5706	sshd: insecure connection attempt (scan).	recon, syslog, sshd	PCI_DSS GPG13 GDPR NIST_800_53 TSC MITRE	6	0095-sshd_rules.xml	ruleset/rules
[illegible]	[illegible]	authentication_fail [illegible]	PCI_DSS GPG13 HIPAA GDPR	[illegible]	[illegible]	[illegible]
5715	sshd: authentication success.	authentication_success, syslog, sshd	PCI_DSS GPG13 HIPAA GDPR NIST_800_53 TSC MITRE	3	0095-sshd_rules.xml	ruleset/rules
5760	sshd: authentication failed.	authentication_failed, syslog, sshd	PCI_DSS GPG13 HIPAA GDPR NIST_800_53 TSC MITRE	5	0095-sshd_rules.xml	ruleset/rules
11512	MS-FTP: Multiple FTP errors from same source.	syslog, msftp	PCI_DSS HIPAA GDPR NIST_800_53 TSC MITRE	10	0190-ms_ftpd_rules.xml	ruleset/rules
80606	Attempt to connect to screen sharing with username **dstuser** from **ip_address** failed.	authentication_failed, invalid_login, macOS	PCI_DSS GPG13 HIPAA GDPR NIST_800_53 TSC MITRE	5	0960-macos_rules.xml	ruleset/rules
80607	Attempt to connect to screen sharing with username **dstuser** from **ip_address** succeeded.	authentication_success, macOS	PCI_DSS GPG13 HIPAA GDPR NIST_800_53 TSC MITRE	3	0960-macos_rules.xml	ruleset/rules
91823	Powershell script used "Invoke-command" cmdlet to execute code on remote computer	windows, powershell	MITRE	14	0915-win-powershell_rules.xml	ruleset/rules
92020	A file was copied to other system over SSH using pscp.exe	sysmon, sysmon_eid1_detections, windows	MITRE	6	0800-sysmon_id_1.xml	ruleset/rules
92105	Possible suspicious access to Windows admin shares	sysmon, sysmon_eid3_detections, windows	MITRE	3	0810-sysmon_id_3.xml	ruleset/rules

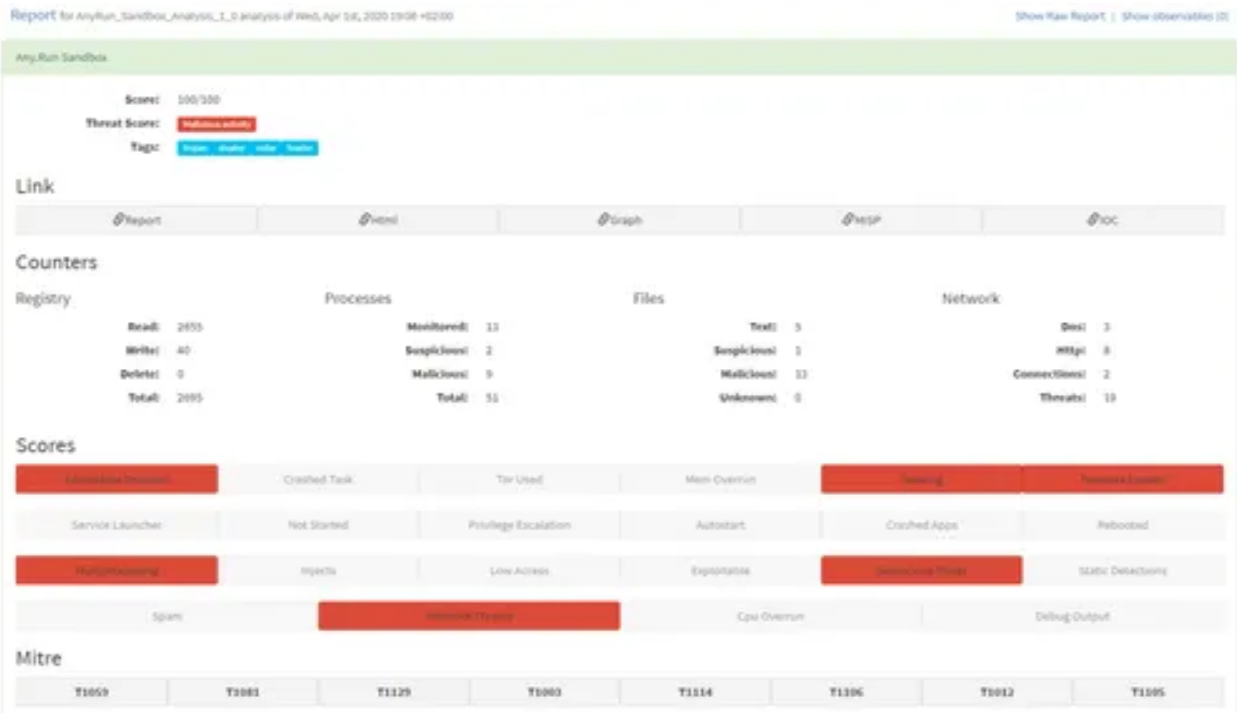
Analysis: The screenshot displays Wazuh SIEM rule management where security rules detect suspicious activities. These rules are mapped to security standards and MITRE ATT&CK; techniques to help SOC teams automatically detect threats such as unauthorized logins and suspicious system activity.

Figure 5: CrowdSec Firewall Blocking



Analysis: This screenshot shows firewall logs where malicious IP addresses are blocked. CrowdSec automatically blocks suspicious traffic attempting to connect to protected systems, preventing attackers from accessing internal resources.

Figure 6: Malware Sandbox Analysis Report



Analysis: This screenshot represents a malware analysis report generated by a sandbox environment. The report shows indicators such as threat score, process activity, registry modifications, and network connections. Such analysis helps analysts confirm malicious behavior in

suspicious files.

Conclusion

This report demonstrates practical SOC investigation tasks including evidence collection, memory forensics, vulnerability analysis, SIEM rule monitoring, firewall blocking, and malware analysis. These activities represent key processes used by Security Operations Centers to detect, investigate, and respond to cyber threats effectively.